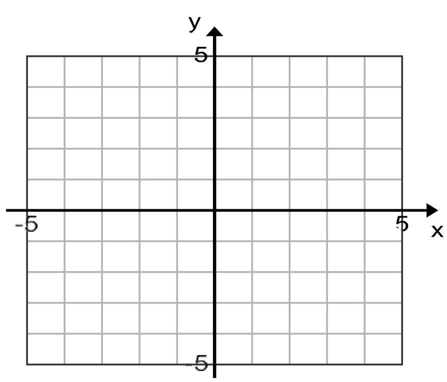
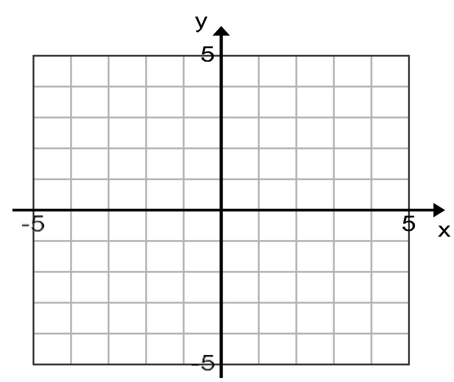
HOMEWORK DAY 6 – *Basic Graph and Transformation (Chapter 6 Section 6.1)*

1. §6.1 55

2. (Follow the instructions as we did in class) Graph $f(x) = -(x+2)^2 + 1$ by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve.

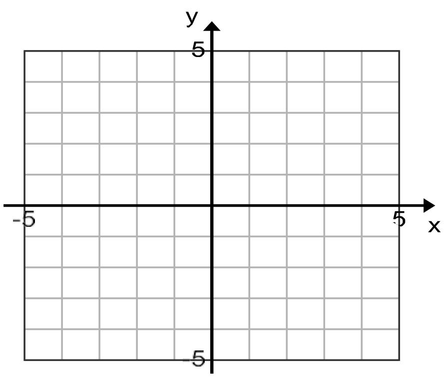
Basic curve	Stretch/Shrink	Reflection: ___	Shifts: ___________
$y =$ _____	$y =$ ______	$y =$ _______	$y = -(x+2)^2 + 1$

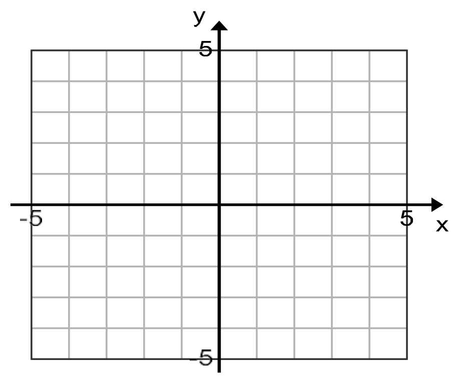




3. (Follow the instructions as we did in class) Graph $f(x) = 2\sqrt{x+3} - 1$ by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve.

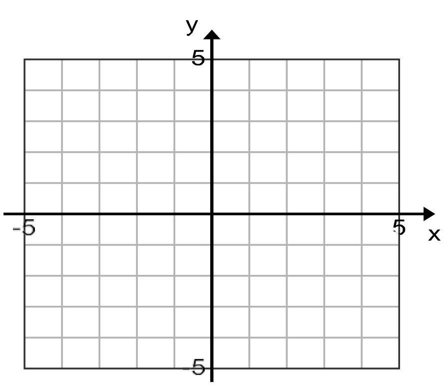
Basic curve	Stretch/Shrink	Reflection: ___	Shifts: ___________
$y =$ _____	$y =$ ______	$y =$ _______	$y = 2\sqrt{x+3} - 1$

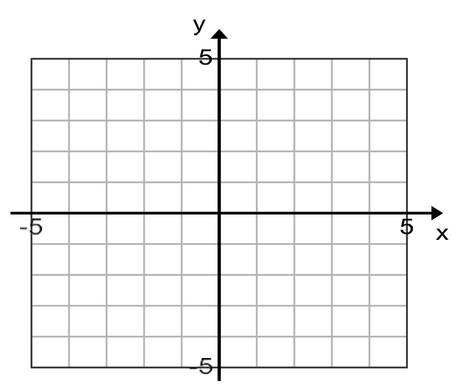




4. (Follow the instructions as we did in class) Graph $f(x) = \sqrt{-x+3}$ by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve.

Basic curve	Stretch/Shrink	Reflection: ___	Shifts: ___________
$y =$ _____	$y =$ ______	$y =$ _______	$y = \sqrt{-x+3}$





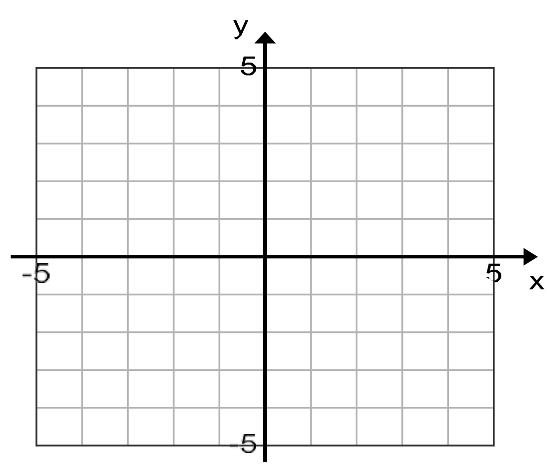
5. §6.2 47

6. Graph $f(x) = \begin{cases} 3, & -4 < x < -1 \\ -x, & -1 \le x < 3 \\ \sqrt{x-3}, & x \ge 3 \end{cases}$

(a) Evaluate $f(-5) =$ $f(0) =$ $f(-1) =$ $f(3) =$

(b) Graph $f(x) = 3,\ -4 < x < -1$

(e) Now combine all three and put them togehter



(c) Graph $f(x) = -x,\ -1 \le x < 3$

(d) Graph $f(x) = \sqrt{x-3},\ x \ge 3$

7. §6.3 30

8. Answer the following by using the average rate of change formula on the given interval.

(a) Determine the ARC of $f(x) = x^2 - 3$ on the interval [1,3].

(b) Determine the ARC of $f(x) = \sqrt{x-1}$ on the interval [2,5].

9. §6.3 35

10. §6.3 40

11. Let $f(x) = \frac{1}{x+2}$. Evaluate and simplify the following

(a) $f(x+h)$

(b) $\frac{f(x+h)-f(x)}{h}$